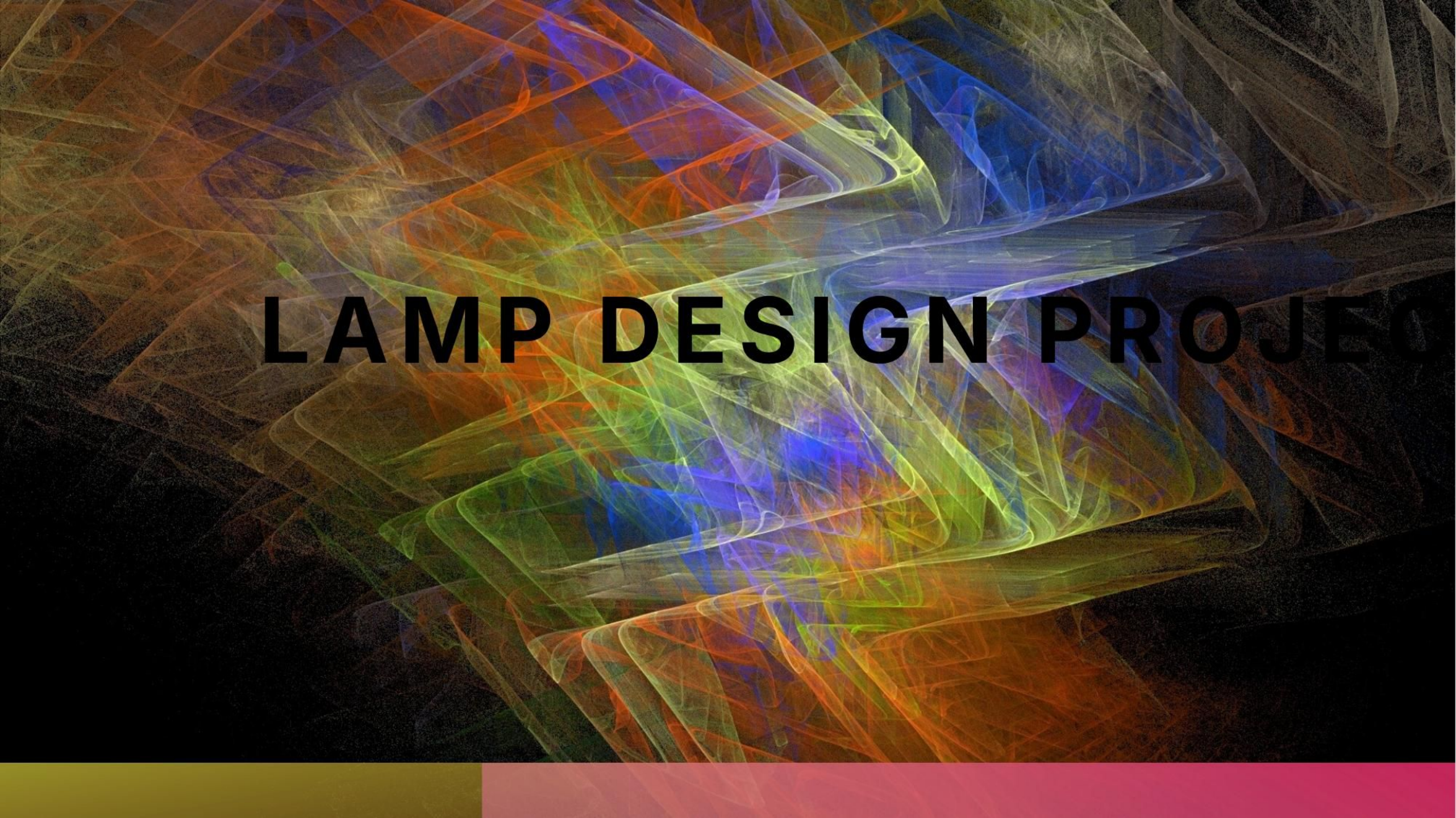
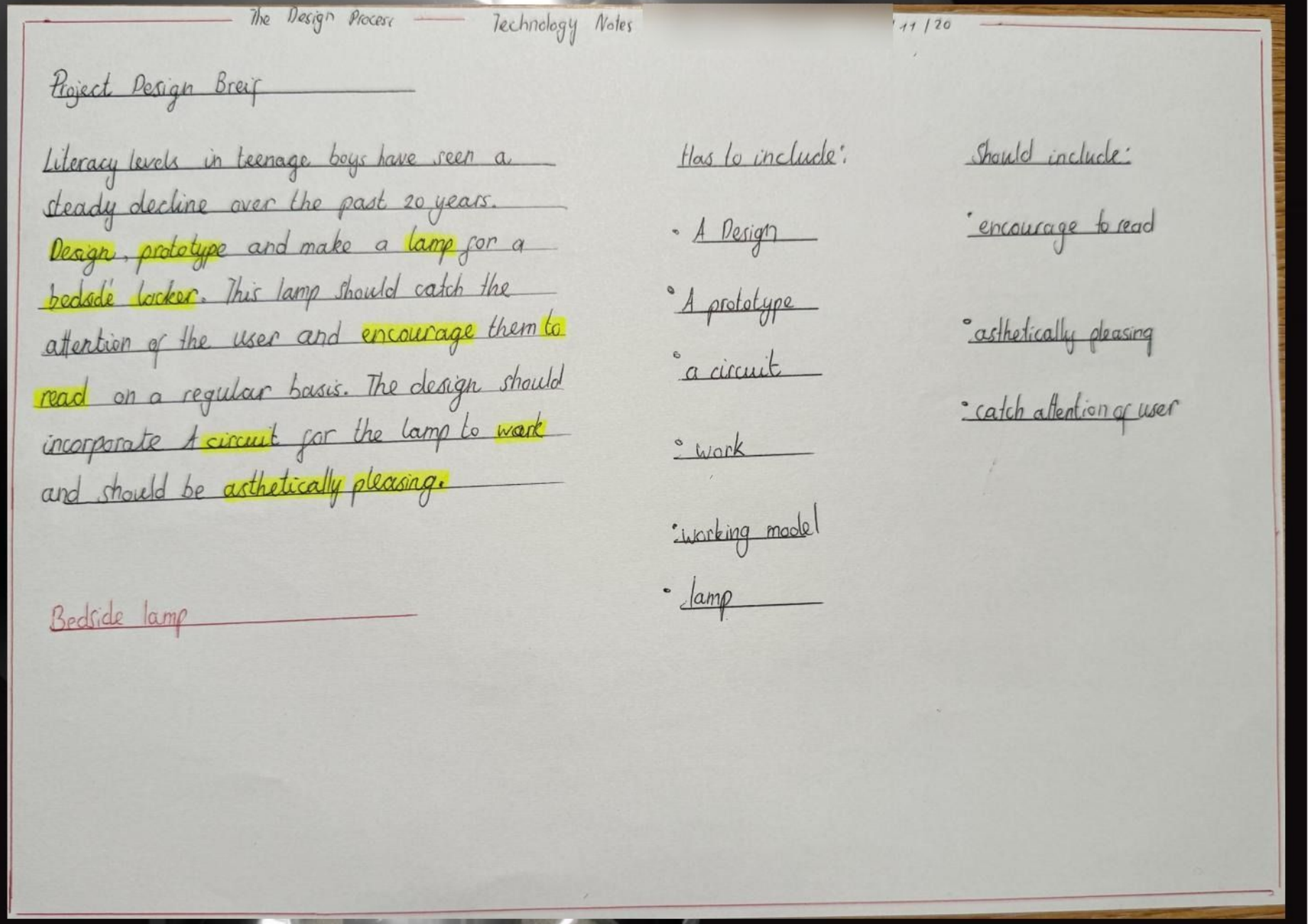

LAMP DESIGN PROJE

# DESIGN BRIEF

For the design brief I analyzed the brief by highlighting the most important parts. Then listing what is mandatory and imperative and what is non mandatory, thus making it easier to understand the project specifications.



The Design Process — Technology Notes — 11/20

Project Design Brief

Literacy levels in teenage boys have seen a steady decline over the past 20 years. Design, prototype and make a lamp for a bedside locker. This lamp should catch the attention of the user and encourage them to read on a regular basis. The design should incorporate A circuit for the lamp to work and should be asthetically pleasing.

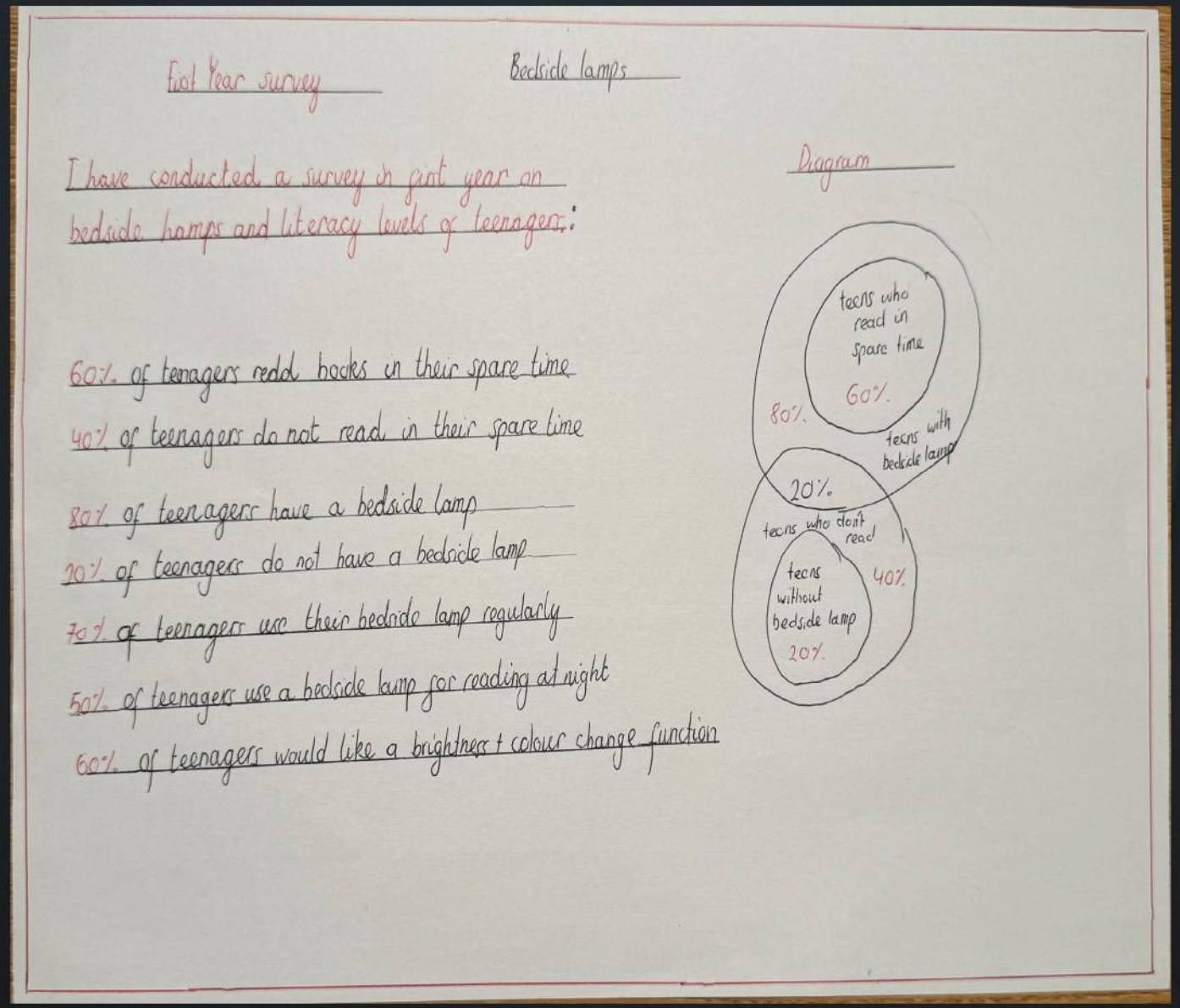
Has to include:

- A Design
- A prototype
- a circuit
- work
- working model
- lamp

Should include:

- encourage to read
- asthetically pleasing
- catch attention of user

Bedside lamp

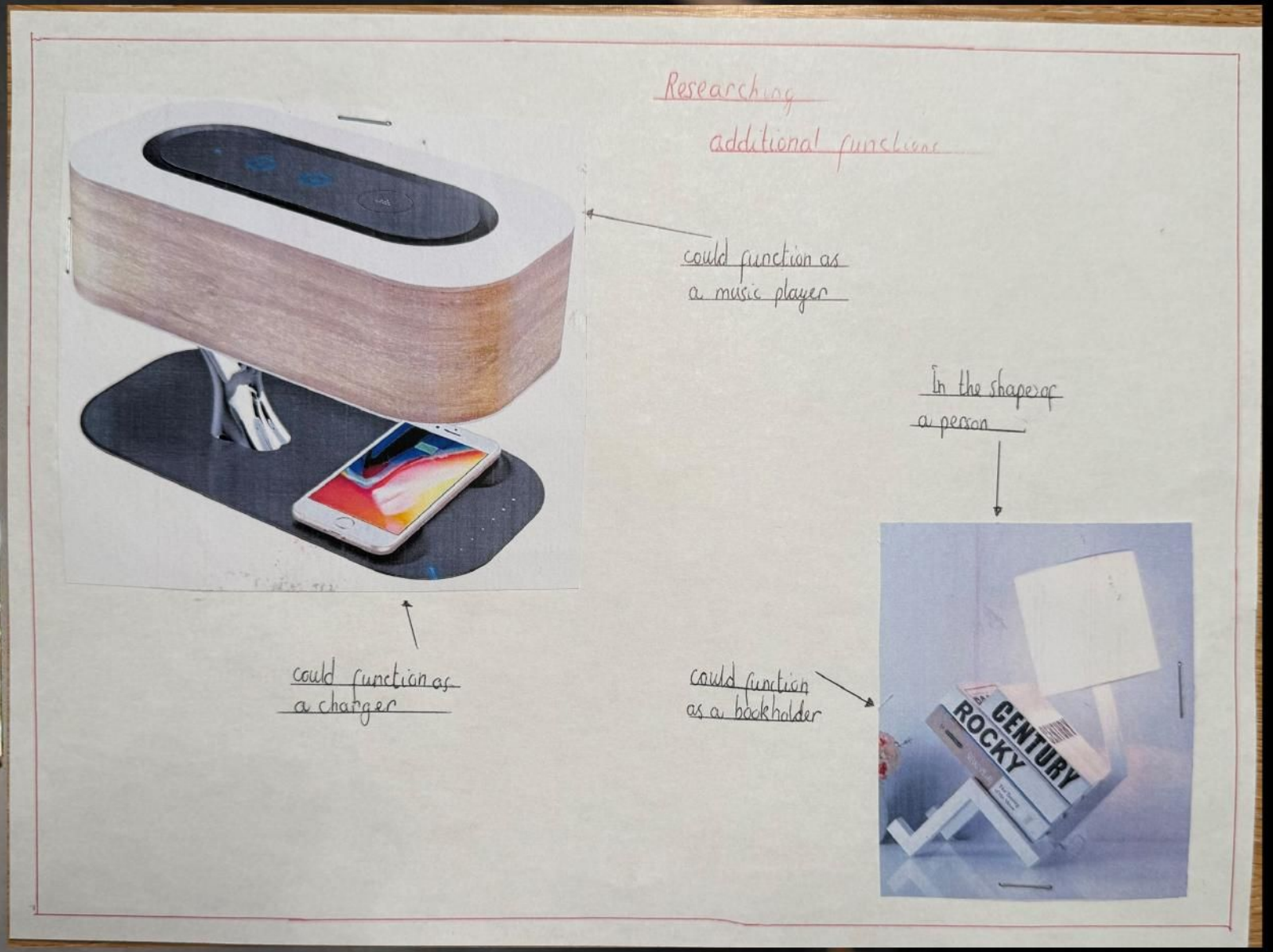



# SURVEY

In order to figure out what to design I started by conducting a survey in first year. This was to understand why teenagers buy lamps and what designs and functions they would be interested in. I questioned many people and found the information shown on the left.

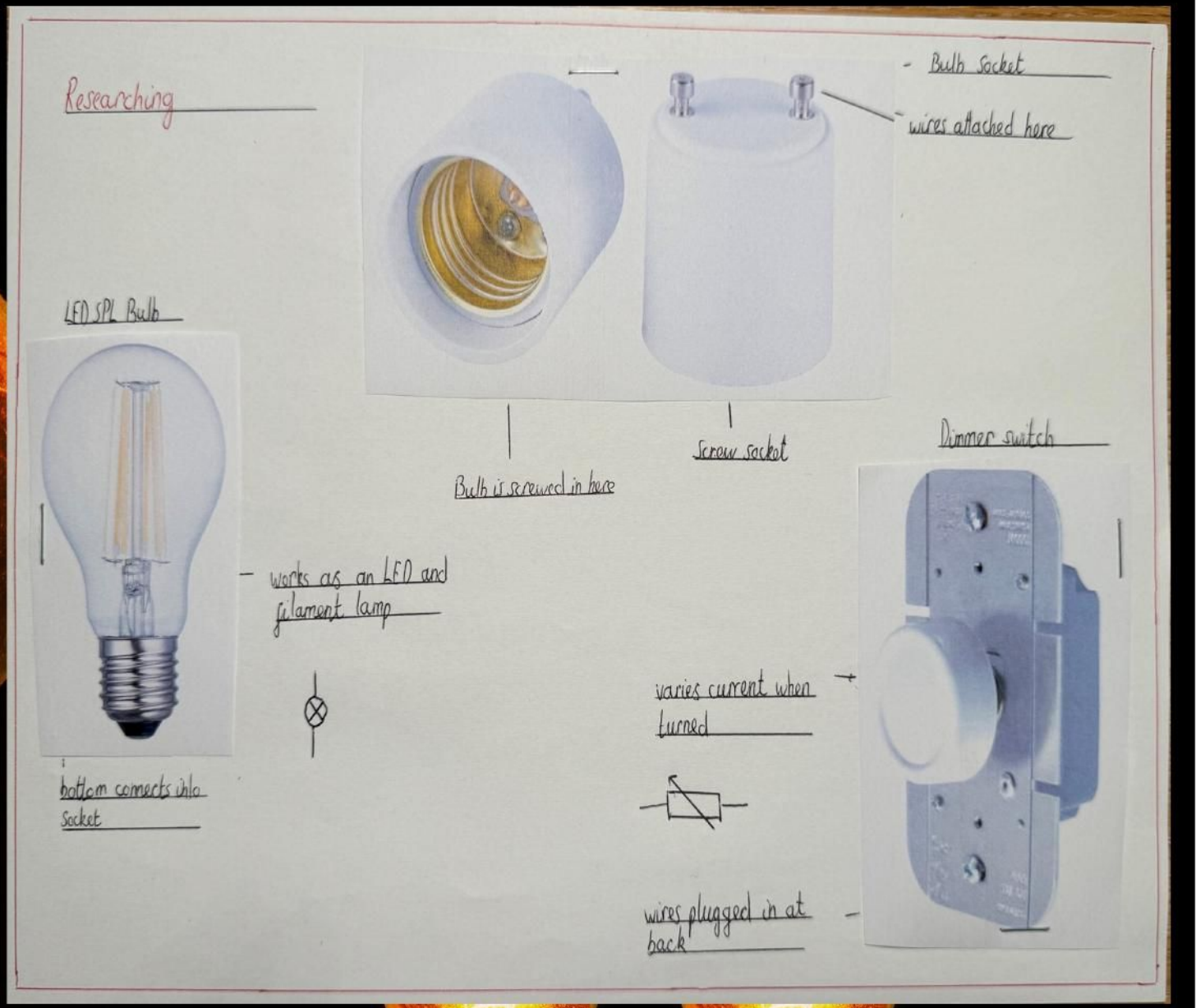
# RESEARCH

While researching I decided to find inspiration for my lamp and for any additional functions by finding unusual lamp designs. Through doing this I also found out a lot about the functions of these lamps.



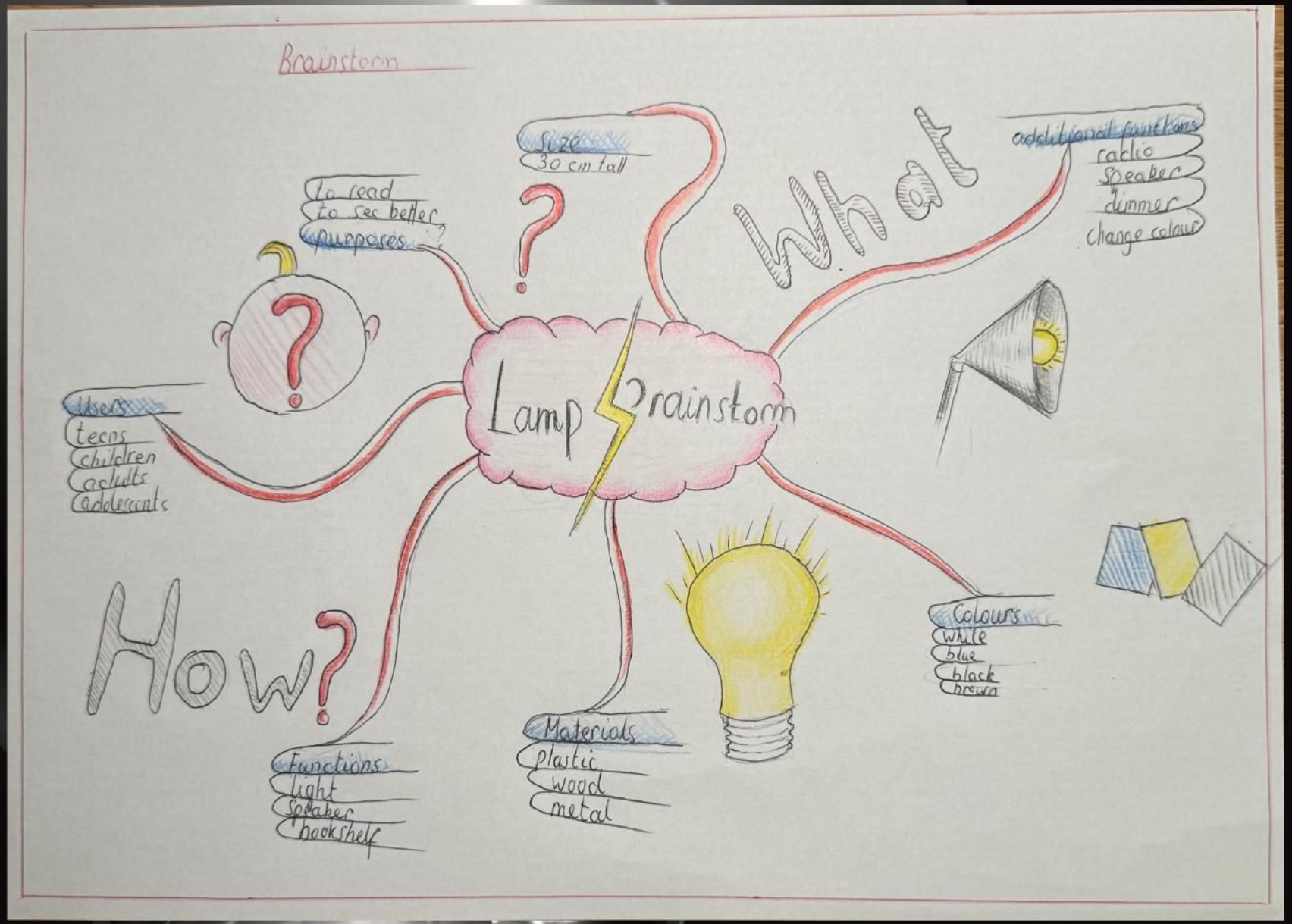
# RESEARCH

To understand how the basic components of a lamp work I spent time researching how bulb sockets, led's and other such components work. This page is a general conclusion my results.



# MINDMAP

For my mindmap I summarized all of my ideas on the topic of lamps, and wrote it down in a coulorful and expressive way.



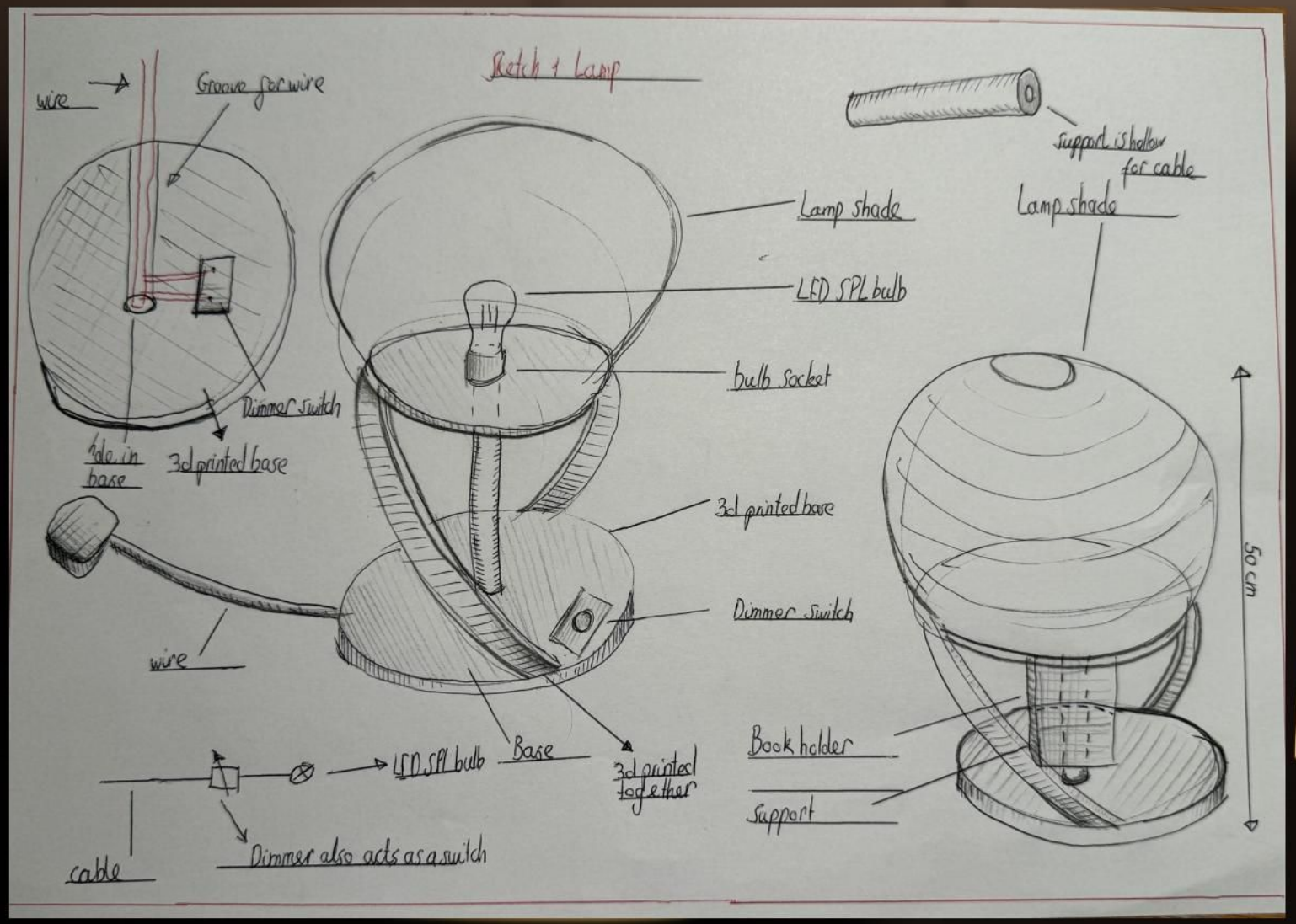
# MOOD BOARD




For my mood Board I expressed all of my ideas using images (mostly inspired by nature).

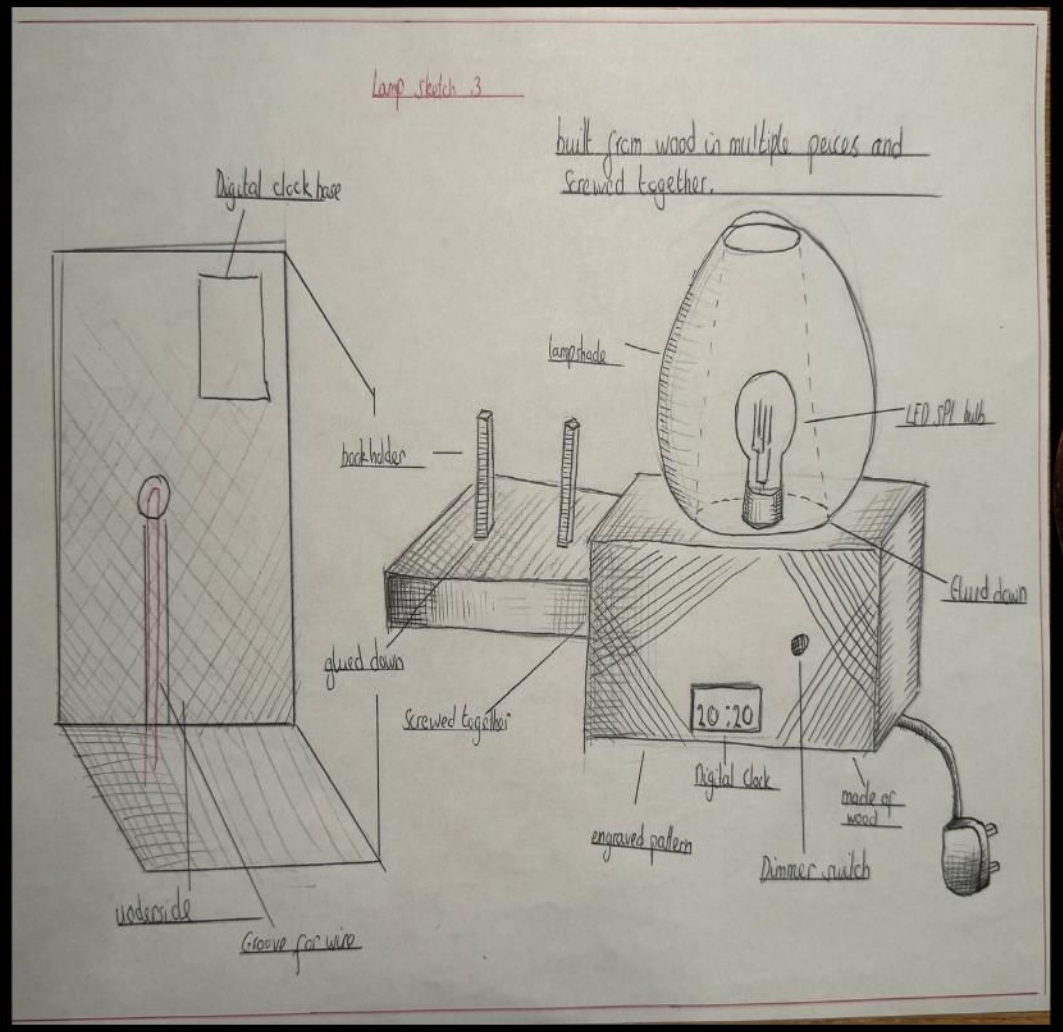
I later used the ideas I received from these images to desi[illegible] my lamp.

# DESIGN SKETCHES




For my first sketch I was inspired by an air-balloon competition I was once at. I created a dome surrounding the light bulb and a base (representing the basket of an air- balloon). The bases are 3d printed and are made to act as supports for books.

# DESIGN SKETCH

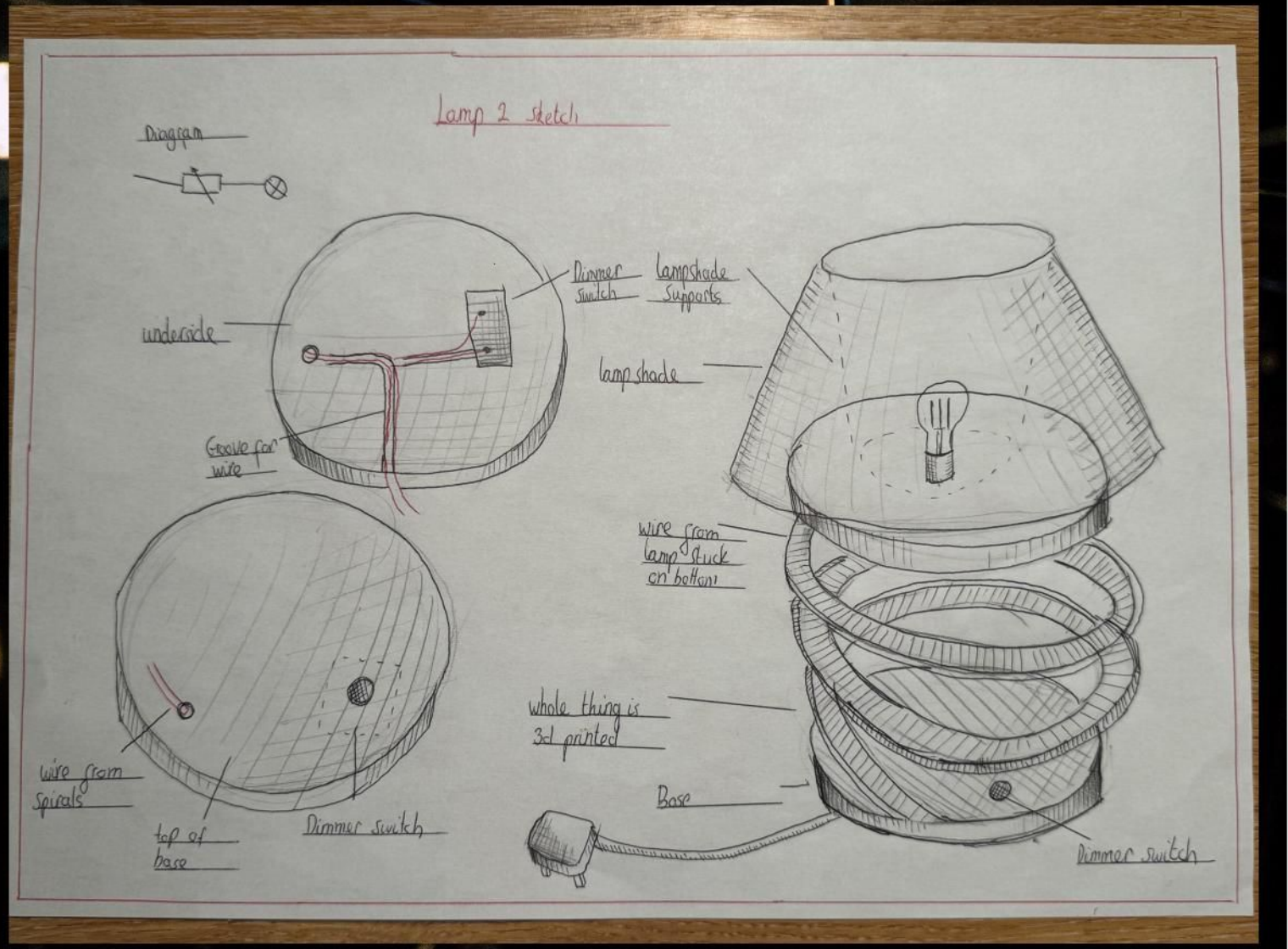




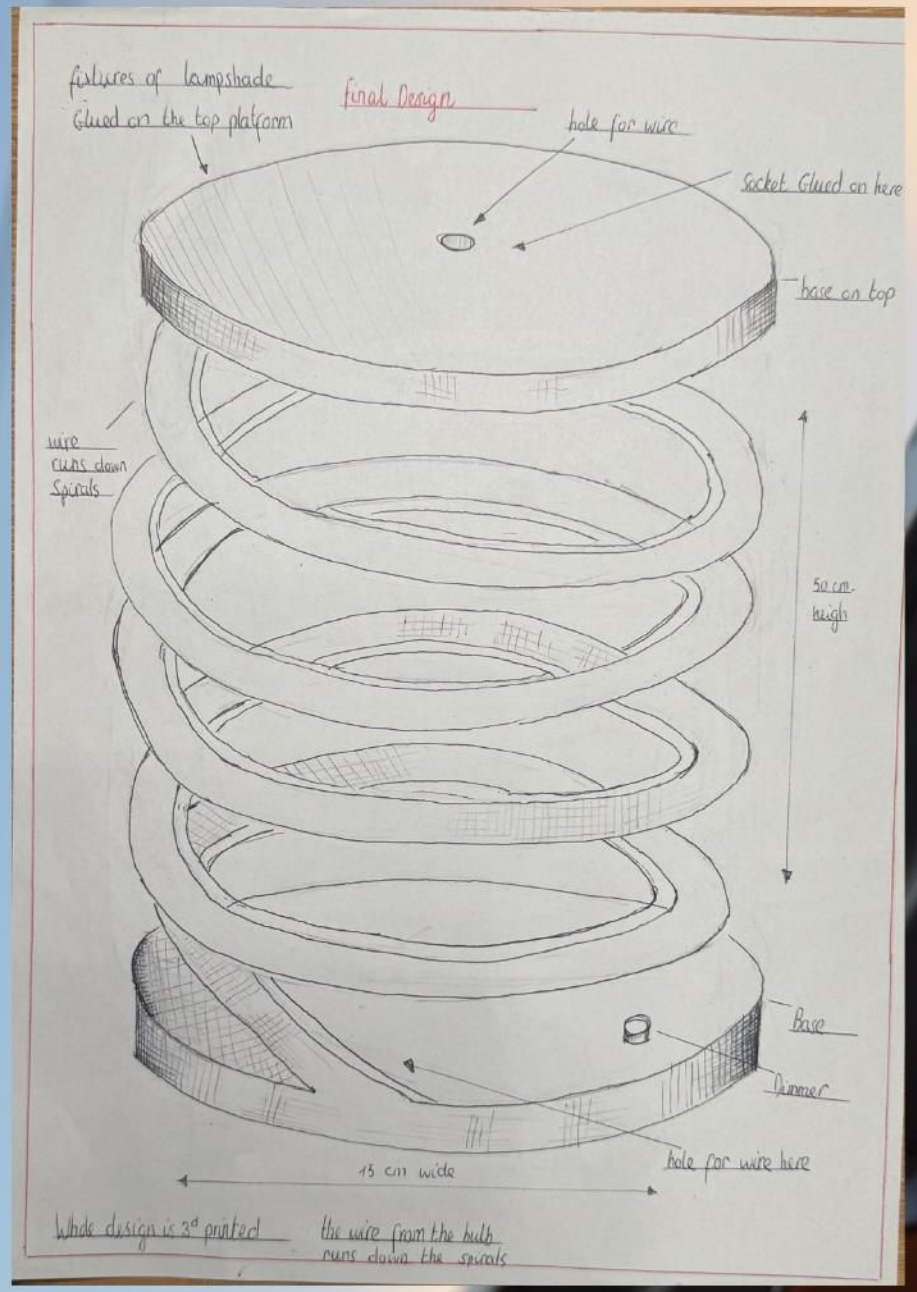

In my second design Sketch I was inspired by the domed houses I had seen on Skellig Island while watching a star wars movie.

# DESIGN SKETCHES

For my third design I was inspired by the suspension in a car. This drove me to design a lamp similar to the one shown here.



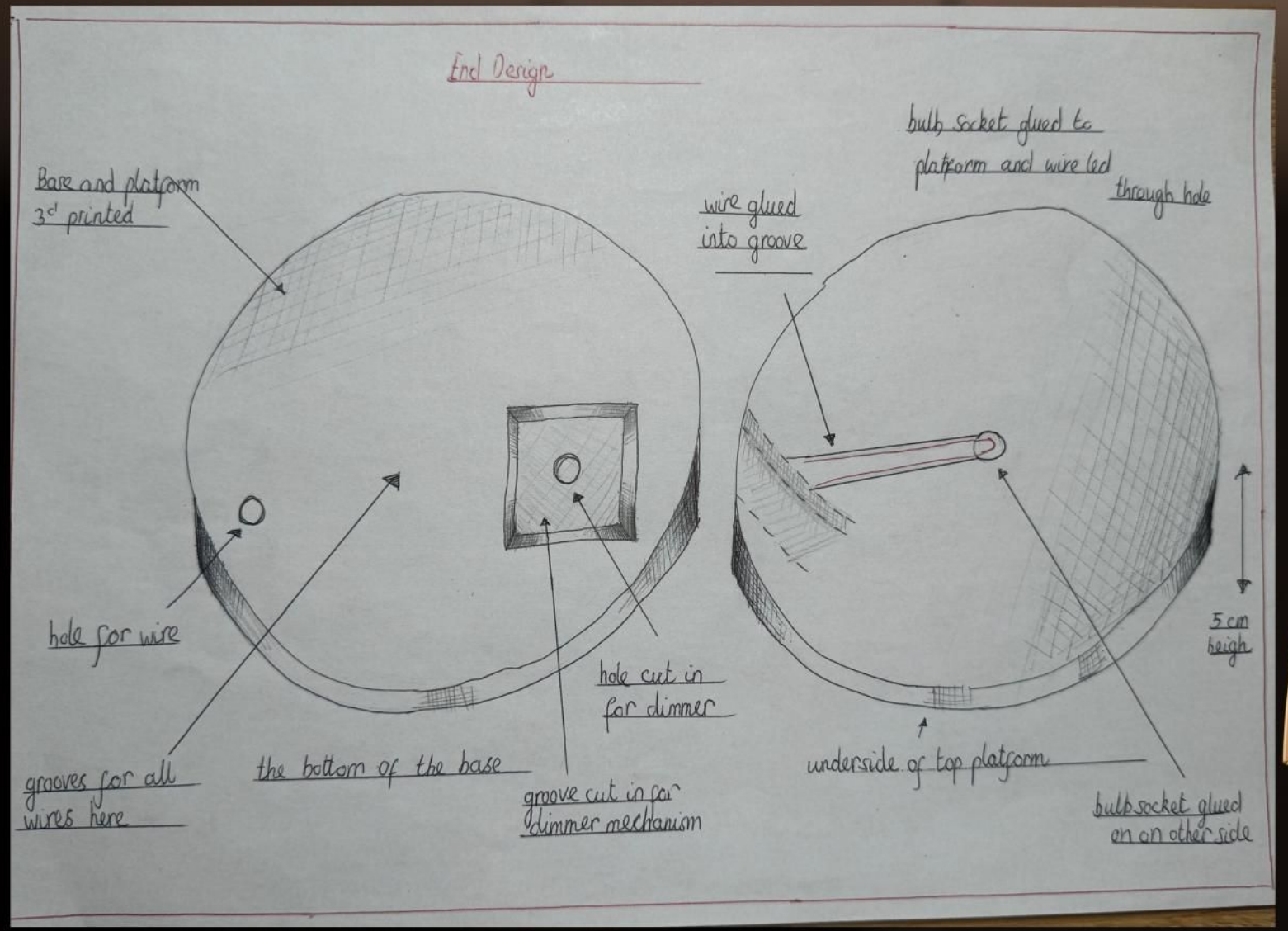
# FINAL DESIGN




In this sketch I sketched the bases of the lamp. The spiral is inspired by my mood board and is 3d printed together with the base. A dimmer switch is also installed in the base.

# FINAL DESIGN

## Base




In the sketches is depicted the bases on the top of the Lamp and the bottom. The Wiring is shown, and Hollows in the plastic Depict where the circuitry Is going to be wired( Diagram not fully accurate)

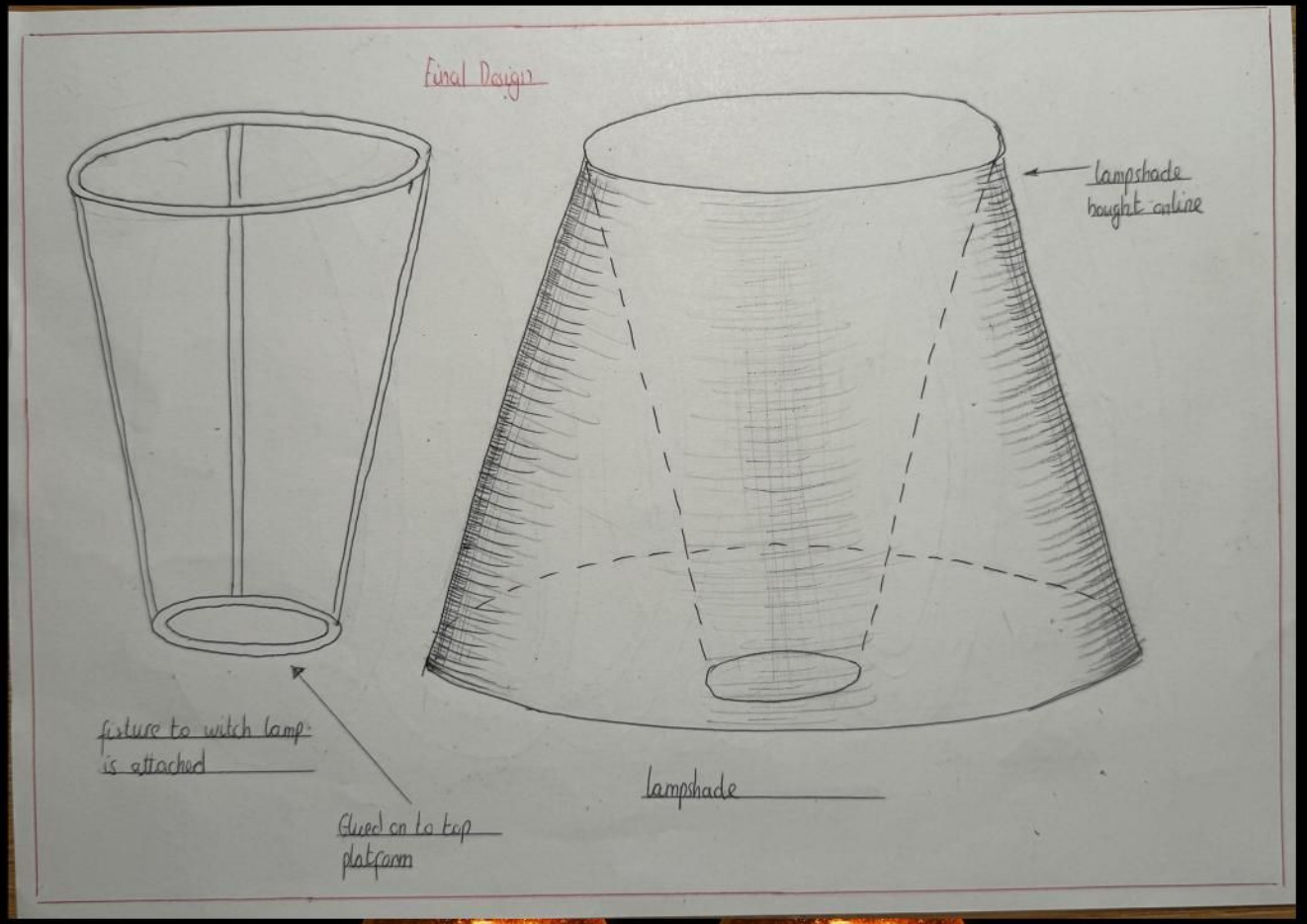
# FINAL DESIGN

Lampshade and supports

I will buy a lampshade and the supports of AliExpress.

Lampshade = 5.50 euros

Supports/fixture = included in lampshade



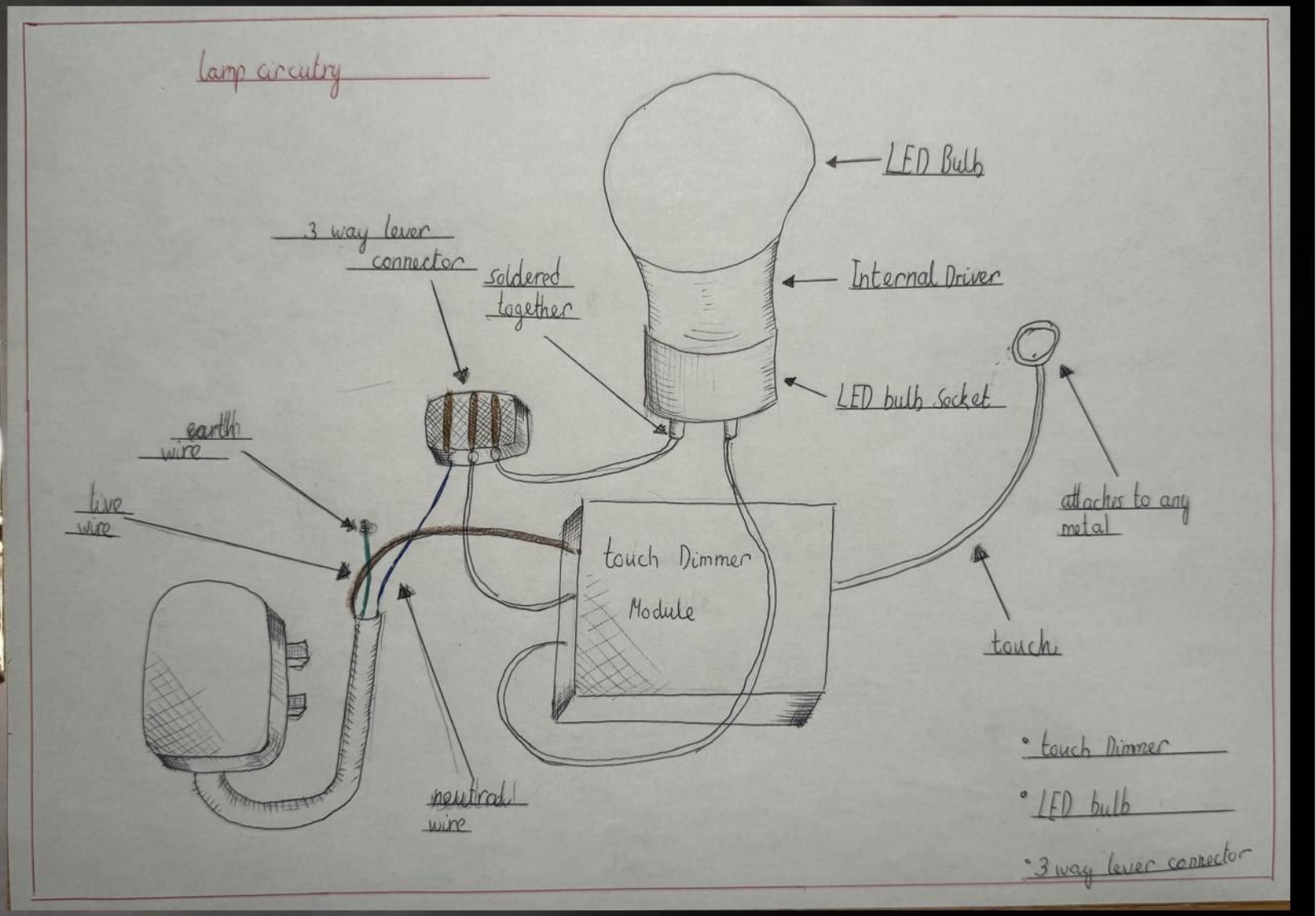
# FINAL DESIGN

## Circuitry

This is the circuitry for my lamp. I will use a touch dimmer module (connected to metal on the lamps base). These items will be ordered of AliExpress.

- Touch dimmer module = 3 euros
- Led bulb = 1.25 euros
- E27 screw socket = 2.41 euros

(delivery price included)



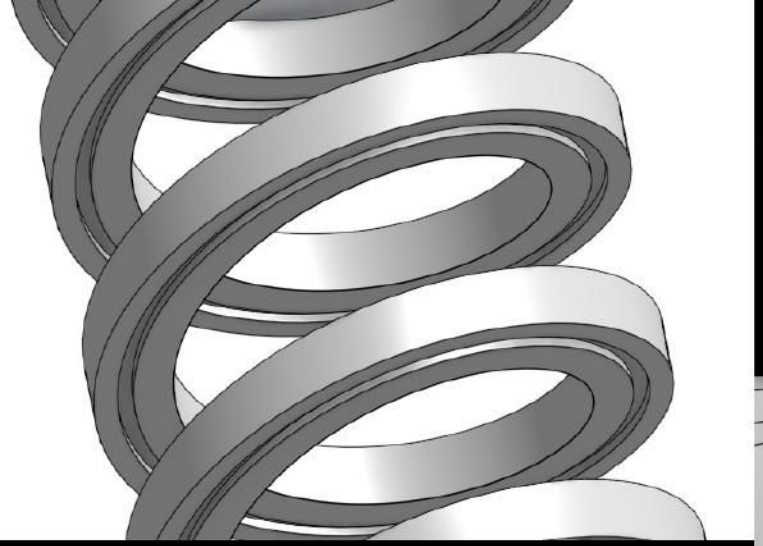


Fig.1

# CAD DESIGN

I designed my project
using CAD.

Groove for wire leading
Up to the bulb on the
Top(fig.3).

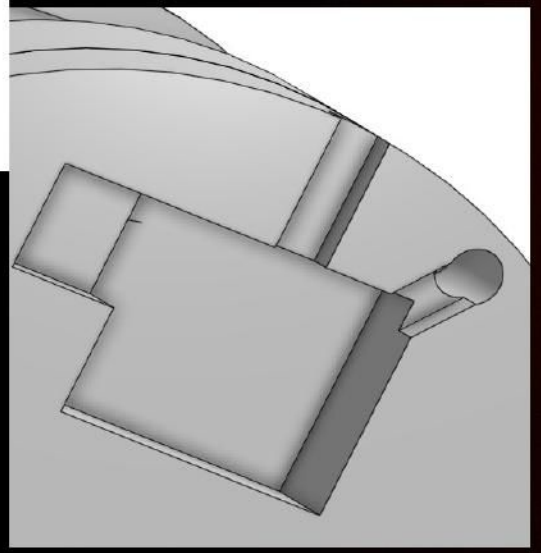


Fig.2

**Hole on top designed for wires leading Away from The screw-Socket.**

Plastic hollowed out
For components (last page).

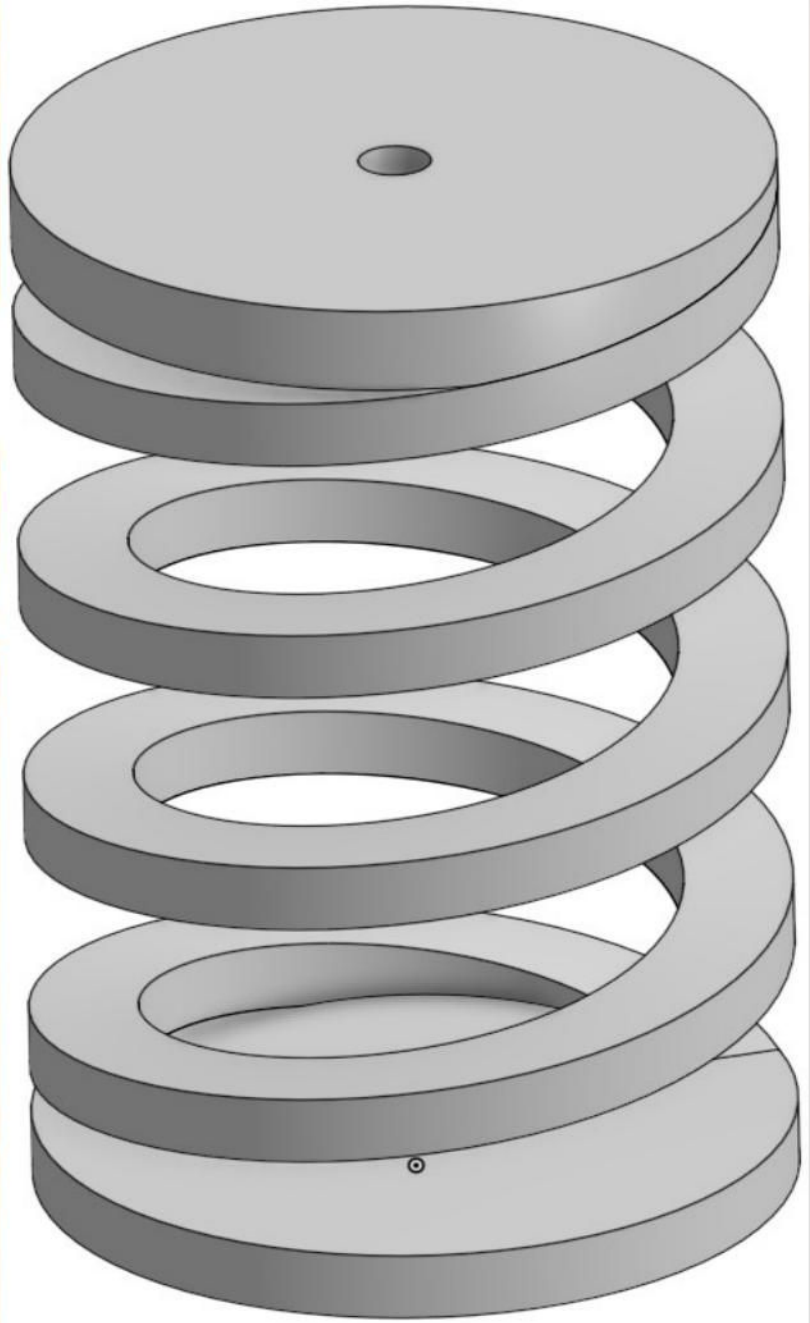


Fig.3

# CAD DESIGN

Interactive 3d models



Models with lampshade

# SURVEY CONCLUSION

In conclusion of the survey conducted at the start of my project

The majority of teenagers would have liked touch dimmers and LED’s installed in their lamps. I have incorporated both of these in my project.



# PROJECT CONCLUSION

- Does it fit specifications ? (brief)
- Is your design aesthetically pleasing ?
- Does it work ?(stability, lighting)
- Ergonomics ?

The lamp Is aesthetically pleasing and thus will entice users to read.

The design is aesthetically pleasing as it is shaped in a spiral.

The lamp is ergonomically built so that no wires are visible and the lamp controls are simple and comfortable.